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Prompt Engineering

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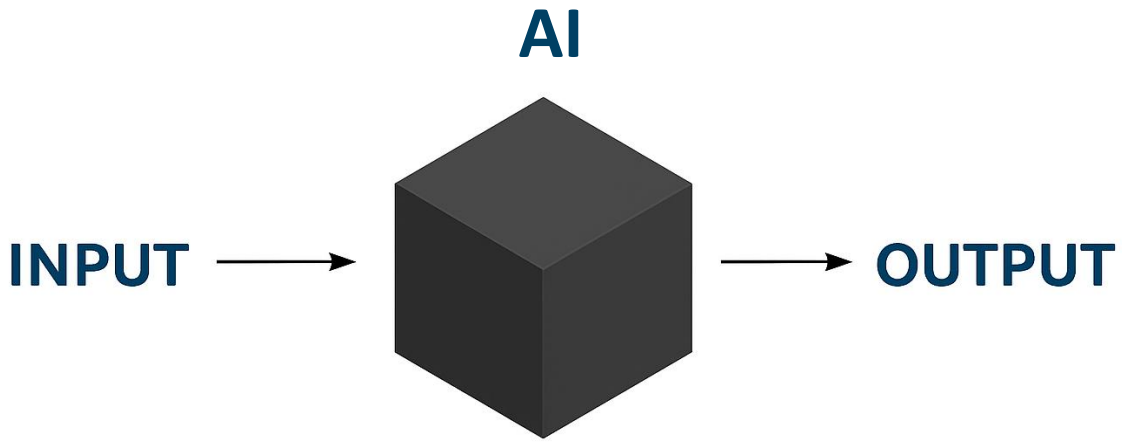
Overview

- Intro to GenAI & LLM
- What & why prompt engineering
- Common behaviors of LLM
- Key prompting techniques (patterns + examples)
- Summary



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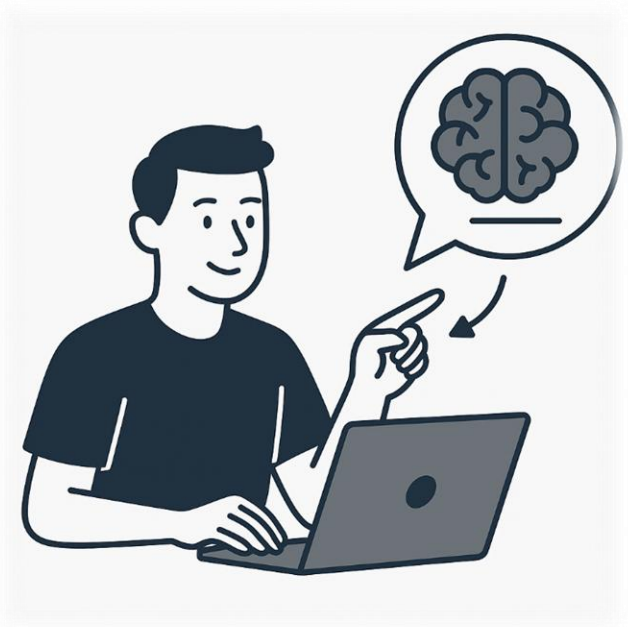
Warm-Up: Your Experience With GenAI



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- What GenAI tools do you use most often?
- For what kinds of tasks do you use them?
- How do you usually interact with them?
- How much do you rely on them for different tasks?
- What problems or frustrations have you encountered?
- What lessons or best practices have you learned from using GenAI?

Warm-Up: Let’s Try It

Image you are the instructor and want to design a quick task that hook students into today’s topic – prompt engineering, and helps them experience how GenAI behaves before learning any prompt engineering.

Do it now:

- Create your task
- Use GenAI to complete it the way you normally would
- Share your output and how you prompted

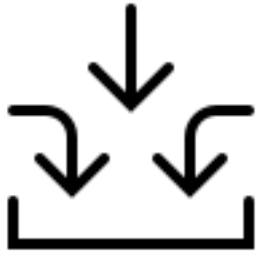
Reference

- Jules White, *Prompt engineering for ChatGPT*, Coursera.
- Google Cloud, *What is prompt engineering?*
- DAIR.AI, *Prompt engineering guide*.
- Other Courses
  - IBM, *Generative AI fundamentals*, Coursera.
  - IBM, *Generative AI: Prompt engineering basics*, Coursera.
  - Google, *Google prompting essentials*, Coursera.
  - Google, *Start writing prompts like a pro*, Coursera.
  - AWS, *Essentials of prompting engineering*, Coursera.
  - Jules White and Bob Higgins, *Generative AI leadership & strategy specialization*, Coursera.

Intro to GenAI & LLM

- LLMs generate text by predicting the next word based on patterns learned from massive datasets.
- Each new word becomes part of the input, so the model updates its response step by step.
- LLMs do not “understand” like humans — they follow statistical patterns, not reasoning.
- They continue generating until they reach an ending that seems complete.
- Why this matters
  - LLMs are powerful, but their **output depends heavily on how you prompt them**, which is why prompt engineering is essential.

What Is a Prompt



- A prompt is the **input** you give a large language model — your *instructions*, *question*, *text, image, video, even a single character*.
- It tells the model what you want, how you want it, and what constraints to follow.
- Because LLMs generate text by predicting the next word, the prompt shapes the region of language the model explores.
- A clear prompt reduces ambiguity and guides the model toward more accurate, relevant, and useful output.

→ Key idea: A prompt is ***not just a question*** — it's a way of steering the model's behavior.

Intuition Behind Prompts

- LLMs follow patterns in your prompt. They try to continue the “story” or style you start.
- The more specific the prompt, the more the model can align with your intent, tone, and structure.
- The prompt acts like a map, narrowing the model’s choices among many possible next words.
- By adjusting **wording**, **roles**, **examples**, or **structure**, you shift the model into a different (and often better) part of its probability space.

→ Key idea: Good prompts intentionally guide the model toward the kind of output you want — clearer, deeper, more accurate, and better aligned with your goals.

Common Behaviors of LLM

- Randomness
  - Outputs can vary each time for the **same prompt**
- Inconsistency
  - Tone, depth, and structure may shift without clear instructions.
- Context-dependent
  - Models adapt within a conversation but don’t retain memory **across sessions** permanently.
- Prompt size limitations
  - Very long inputs may be truncated, complex tasks need chunking or summarizing.

→ Key idea: LLMs are powerful but unpredictable unless your prompts provide structure and clarity.

Common Behaviors of LLM

- Built-in root prompts
  - GenAI tools include hidden system rules that enforce safety, tone, and general behavior.
- Your own “root prompt”
  - You can set a guiding instruction at the **start of a session**.
  - *Example: “For this conversation, act as my research assistant: concise, analytical, and always cite assumptions.” → Sets consistent behavior throughout the dialogue.*
- Reusable prompts templates
  - Effective prompts can be reused like templates: roles, structure, and style stay stable even when content changes.

→ Key idea: A clear root prompt makes the model more consistent and reliable.

What Makes a Good Prompt

- A prompt guides the model's behavior
  - **Who to be, what to do, and how to present the answer**.
- Strong prompts typically include
  - Role + Task + Context + Structure + Constraints + Examples
- Why it works:
  - Reduces ambiguity
  - Narrows the model's "language space"
  - Aligns output with your intent, audience, and purpose
- Expert insight: Clarity beats cleverness. Specific, structured prompts consistently outperform short or vague ones.
- Key idea: Good prompts provide **direction**, the model provides **content**.

Revisit the Warm-Up Task

Image you are the instructor and want to design a quick task that hook students into today's topic – prompt engineering, and helps them experience how GenAI behaves before learning any prompt engineering.

Now apply what you've learned so far:

- Rewrite your original prompt using clear roles, structure, context, and constraints.
- Generate a new output with this revised prompt.
- Compare: How is the new output different from your original one?
- Share your new output and the prompt you used.

Prompting Pattern 1 – Persona

- Purpose
  - Guide the model by assigning it a specific role so it adopts the knowledge, tone, and perspective of that persona.
- How to use
  - Act as persona X
  - Perform task Y

A short sentence, rich information

- Examples
  - "Act as a nutritionist. I'll tell you what I ate; you evaluate my choices."
  - *"Act as a gourmet chef. Comment on the ingredients I provide."*
- Why it works
  - A persona narrows the model's "identity," leading to clearer, more relevant, and more consistent output.

Prompting Pattern 2 – Audience Persona

- Purpose
  - Shape how the model explains something by defining who the audience is.
- How to use
  - Explain X to me. Assume I am Persona Y. (e.g., a beginner, an expert, a child, someone from another field)
- Examples
  - Explain large language models to me. Assume I am a bird.
  - Explain how US grocery supply chains work. Assume I am Genghis Khan.
- Why it works
  - The model adapts tone, depth, vocabulary, and analogies to match the target audience—making explanations clearer and more relevant.

Prompting Pattern 3 – Question Refinement

- Purpose
  - Help the model improve your questions so you get clearer, more actionable answers.
- How to use
  - “From now on, whenever I ask a question, suggest a better version of the question.”
  - (Optional) “Ask me if I want to use the improved version before answering.”
- Examples
  - “When I ask about dieting, refine my question to focus on healthy habits. Ask for my first question.”
  - “When I ask who the GOAT is, refine the question to highlight different players’ accomplishments.”
- Why it works
  - Better questions lead to better prompts and better output.

Prompting Pattern 4 – Cognitive Verifier

- Purpose
  - Improve accuracy by having the model break a question into helpful sub-questions before answering.
- How to use
  - When you receive a question, follow these rules:
    - Generate additional questions needed to answer it well.
    - Answer those sub-questions.
    - Combine them into a final, stronger answer.
- Example
  - Recipe: Ask about ingredients and equipment → answer sub-questions → produce a workable recipe.
  - Trip planning: Ask about budget, activities, transportation → integrate answers → build a better itinerary.
- Why it works
  - Forces the model to think more completely—reducing mistakes and improving reasoning.

Prompting Pattern 5 – Flipped Interaction

- Purpose
  - Let the model drive the conversation by asking you questions to reach a goal.
- How to use
  - Ask me questions to achieve X.
  - Continue until condition Y is met.
  - (Optional) specify pace: one question at a time, two at a time, etc.
- Examples
  - Ask me questions to create variations of my marketing materials. Continue until you understand my draft, audience, and goals. Ask the first question.
- Why it works
  - The model collects targeted information first—leading to more personalized and accurate results.

Prompting Pattern 6 – Few-Shot Examples

- Purpose
  - Show the model the style, format, or reasoning you want—so it imitates your examples.
- How to use
  - Provide a few examples of the desired input → output pattern
  - Then give a new input for the model to continue in the same style
  - Works well for tone, structure, reasoning steps, and action patterns
- Simple Example:
  - Input: “Explain inflation in one sentence.”
  - Output: “Inflation is when prices rise and money buys less.”
  - Input: “Explain supply chain disruption in one sentence.”

Why it works

- LLMs are pattern-matchers—examples create the pattern you want them to follow.

Many More Patterns Exist… And That’s the Point

- We’ve covered several core prompt patterns but there are many others.
- You don’t need to memorize them all.
- What matters is learning to think creatively about how to guide the model.
- Key ideas
  - Use patterns as inspiration, not rules.
  - Treat prompting as a dialogue — iterate, refine, and explore.
  - Experiment with roles, structure, constraints, questions, and examples.
  - Develop your own “prompt style” that consistently gives you the outcomes you want.
- Goal
  - Find your way of prompting—one that reliably produces high-quality, meaningful results.

Warm-Up Task: Now Apply the Patterns

Image you are the instructor and want to design a quick task that hook students into today's topic – prompt engineering, and helps them experience how GenAI behaves before learning any prompt engineering.

- Rewrite your original prompt using personas, refined questions, audience personas, few-shot examples, or any other pattern.
- Generate a new output with your improved prompt.
- Compare: How does this version differ from your **first, second** attempt?
- Share your new prompt and output.
- Goal:
  - See how intentional prompt patterns transform GenAI from "OK output" to high-quality, customized results.

Summary

- Output from LLMs rely on prompts — clear inputs lead to clearer outputs.
- Effective prompts include role, task, context, structure, and constraints.
- LLMs show randomness, inconsistency, and context dependence — prompting helps control this.
- Several patterns
  - Persona, question refinement, cognitive verifier, audience persona, flipped interaction, few-shot examples
- Prompt engineering is creative and iterative — a dialogue, not a one-shot request.
- Improved prompts → higher-quality, more useful outputs.

→ Key takeaway: Better prompts make GenAI a more reliable, powerful partner in your learning and decision-making.